

Chikungunya adolescent vaccination and the distinct risk–benefit landscapes



Chikungunya virus is expanding globally, with outbreaks reported across Africa, Asia, and the Americas, and locally acquired infections documented in Europe.¹ To date, 104 countries have reported transmission, placing an estimated 2.8 billion people at risk.² The Americas have borne the greatest burden, with about 3.7 million symptomatic cases notified between 2013 and 2023.³

Clinically, chikungunya is characterised by acute fever and severe, often disabling, arthralgia. Although symptoms usually resolve within 2 weeks, persistent joint pain develops in about approximately 30–60% of patients for 3 months, and up to 44% experience symptoms lasting for years.⁴ Mortality increases three-fold to six-fold during the first 84 days after infection,³ particularly among older adults, men, individuals of African ancestry, or those with comorbidities. Adolescents aged 10–19 years have a 1.56-fold (95% CI 1.49–1.63) higher risk of hospitalisation than the population average.^{5,6} Thus, vaccines that protect adolescents against chikungunya virus infection and disease are a pressing need.

Two vaccines have received regulatory approval in several countries: the virus-like particle vaccine PXVX0317 (Vimkunya)⁷ and the attenuated virus vaccine VLA1553 (IXCHIQ),⁸ for use in both travellers and residents of endemic countries. The Vimkunya virus-like particle vaccine, which contains no genetic material and cannot replicate, was shown in a phase 3 trial in the USA to be highly immunogenic with a favorable safety profile.⁷ In this issue of the *Lancet Infectious Diseases*, Vera Buerger and colleagues provide the final results of a phase 3 trial of VLA1553 in Brazilian adolescents.⁹

The safety and immunogenicity of VLA1553 were first demonstrated in adults 18 years and older, predominantly White (approximately 80%), a population generally associated with milder chikungunya outcomes.¹⁰ That phase 3 trial enrolled 4128 participants (3093 assigned to the vaccine and 1035 to placebo). Among vaccine recipients, 98.9% achieved protective neutralising antibody titres 28 days post-vaccination,¹⁰ and 96.8% of this cohort remained seropositive after 2 years.⁸ Adverse events were reported by 62.5% of vaccinees and 44.8% of placebo recipients.¹⁰ Building

on those results, Buerger and colleagues conducted a smaller, randomised, placebo-controlled, double-blind, phase 3 trial of VLA1553 in Brazilian adolescents aged from 12 years up to 18 years (502 participants allocated to the vaccine and 252 to placebo).¹¹ The cohort had predominantly mixed ancestry (multiracial, White, and Black or African American), and 18% had previous chikungunya virus infection. Among baseline-seronegative adolescents, 98.8% achieved protective antibody levels 28 days after immunisation.¹¹ However, adverse events occurred in 78.3% of vaccine recipients and 61.5% of placebo recipients during the first 6 months, with one serious vaccine-related event (high fever) and 18 participants (3.6%) who had symptoms of special interest in the vaccine group versus two (1%) in placebo.¹¹ In their current report,⁹ Buerger reports that 98.3% of baseline-seronegative participants remained seropositive after 12 months follow up. In the final analysis, adverse events were reported by 84.7% of vaccine and 76.6% of placebo recipients between 6 and 12 months. Participants with pre-existing chikungunya virus antibodies reported fewer adverse events, and the absence of increased reactogenicity among those with mixed ancestry is regarded as an encouraging.⁹

Despite its demonstrated immunogenicity, on Aug 22, 2025, the US Food and Drug Administration (FDA) suspended the license for the attenuated virus chikungunya virus vaccine VLA1553, citing an unfavourable risk–benefit balance for American travellers to endemic regions.¹² The FDA identified 32 severe adverse events, including three deaths, among 53394 administered doses. Of these, 28 occurred in individuals 50 years and older, most with significant comorbidities. The three fatalities were in patients older than 75 years with severe underlying disease. Among the four younger people with severe adverse events (aged 28–48 years), one had no underlying conditions, while the others had major comorbidities.¹³

In chikungunya virus-endemic regions, tourists from high-income countries typically stay in affluent areas where exposure risk is minimal. During the 2014 World Cup, it was estimated that the risk of dengue virus infection for a 2 week stay in Brazil was less than



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ten per 100 000 travellers (0.01%),¹⁴ and the risk for chikungunya virus infection today is considered to be similar. In contrast, residents living in endemic areas face a mean annual force of infection of 3.1% infection risk of chikungunya virus infection.² Given the vaccine-associated risks, the extremely low infection risk for travellers, and the availability of the non-replicating virus-like particle vaccine, considered safer for older and immunocompromised populations, it is reasonable to favour the virus-like particle vaccine for these groups, while adolescents and young adults living in endemic regions can benefit from either vaccine. **The stark difference in infection risk between travellers and residents underscores a crucial public health distinction: while vaccination offers minimal benefit for travellers, it provides essential protection for populations living in endemic areas.**

We declare no competing interests.

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